

Topic Modeling with Coherence Maximization

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DECLARATION

We hereby certify that the project titled “Optimization Techniques for Machine Learning: Topic Modeling with Coherence Maximization” is a genuine representation of our work, conducted as part of the Project requirement for the CA-CSE275 at Lovely Professional University, Phagwara. This work was carried out under the supervision of Professor Dr. Bhupinder Singh during the period of January to April 2026. All the data and insights presented in this project report are derived from our dedicated efforts and are entirely original.

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CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Project is fit for the submission in Lovely Professional University, Phagwara.

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ABSTRACT

The exponential growth of unstructured text data, particularly in the domain of scientific literature and research articles, necessitates the deployment of automated, robust mechanisms for information retrieval and thematic extraction. Topic modeling, an unsupervised machine learning technique, serves as a cornerstone for discovering latent semantic structures within large corpora. This project explores the optimization of topic models, specifically Latent Dirichlet Allocation (LDA), with a central focus on maximizing topic coherence. Historically, models were optimized using statistical predictive metrics such as perplexity, which often yield results uncorrelated with human judgment. By shifting the optimization objective to maximize the coherence score (specifically the C_v metric), we align the machine learning loss landscape with human interpretability. The ultimate success criterion for this optimization paradigm is the generation of interpretable, semantically distinct, and actionable topics. This report presents a systematic methodology, empirical experimental setup, and quantitative analysis demonstrating that coherence-based optimization leads to vastly superior topic interpretability, solving the disconnect between statistical convergence and human semantic understanding.

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CHAPTER 1 INTRODUCTION

1.1 Problem Statement

In the contemporary landscape of machine learning and natural language processing (NLP), extracting meaningful thematic structures from vast archives of text is a highly complex computational challenge. As researchers, institutions, and enterprises grapple with millions of unstructured research articles, manual categorization becomes practically intractable. While Latent Dirichlet Allocation (LDA) provides a probabilistic framework to cluster word co-occurrences into "topics," extracting *meaningful* and *interpretable* topics remains a substantial hurdle. The primary problem lies in hyperparameter tuning and model evaluation. Unoptimized LDA models frequently generate topics that are overly generic, heavily overlapping, or composed of statistical noise. Furthermore, traditional optimization based on predictive "perplexity" metrics often leads to topics that are mathematically sound but completely incomprehensible to human readers.

1.2 Motivation

The motivation for this study stems from the critical need to bridge the gap between machine learning metrics and human interpretability. A topic model is only as useful as its ability to be understood and acted upon by human analysts. If a model clusters words based on syntactic quirks rather than true semantic relationships, the output is functionally useless for research categorization or literature recommendation. By transitioning from a static application of LDA to an active optimization pipeline focused on "Coherence Maximization," we can fundamentally transform how unsupervised NLP models are trained. Providing a framework that guarantees interpretable, distinct topics will directly benefit researchers attempting to map the evolution of scientific disciplines or build intelligent recommendation systems.

1.3 Objectives of the Study

The primary goal of this study is to address the limitations of traditional topic modeling by implementing a robust optimization pipeline. The specific objectives are:

1. To implement an end-to-end NLP pipeline for text preprocessing, specifically tailored for academic research abstracts.
2. To design an LDA topic modeling architecture and establish a grid-search optimization framework for its core hyperparameters (Number of topics K , α , and β priors).
3. To replace traditional perplexity metrics with the K Coherence Score as the primary objective function for model evaluation.

CHAPTER 2

LITERATURE REVIEW

2.1 Related Work

The evolution of topic model evaluation has been characterized by a paradigm shift from purely statistical measures to semantically driven metrics. Early research on topic modeling, pioneered by Blei et al. (2003), predominantly relied on held-out log-likelihood and *perplexity* to evaluate model fit. However, as topic modeling saw wider adoption, researchers began noticing a disconnect between model performance and human usability.

A seminal study by Chang et al. (2009) revealed a surprising negative correlation between predictive log-likelihood and human interpretability. Topics that achieved the lowest perplexity scores were often deemed the least coherent by human annotators, as the model optimized for complex statistical artifacts rather than meaningful themes. This catalyzed the development of topic coherence metrics by Mimno et al. (2011), who utilized Pointwise Mutual Information (PMI). Later, Röder et al. (2015) introduced a unified framework for topic coherence, demonstrating that the C_v metric achieves the highest correlation with human evaluations.

Table 1: Comparative Analysis of Relevant Studies in Topic Modeling

Author	Year	Method/Focus	Findings
Blei et al.	2003	Latent Dirichlet Allocation (LDA)	Introduced LDA as a generative probabilistic model for discrete data, setting the foundation for modern topic modeling.
Chang et al.	2009	Human Evaluation vs. Perplexity	Revealed a negative correlation between predictive log-likelihood (perplexity) and human interpretability.

Mimno et al.	2011	Semantic Coherence	Introduced automatic
			evaluation metrics using Pointwise Mutual Information (PMI) to identify semantic anomalies.
Röder et al.	2015	C_v Coherence Framework	Demonstrated that the K metric (combining sliding window, NPMI, and cosine similarity) highest correlates with human judgment.
Syed & Spruit	2017	Abstract vs. Full-text analysis	Showed that training LDA on abstracts yields more coherent and dense topics than training on noisy full-text articles.

CHAPTER 3 METHODOLOGY

3.1 Dataset and Preprocessing

The experimental dataset comprises a curated corpus of 10,000 academic research abstracts extracted from computer science and biomedical databases. To prepare the unstructured text for probabilistic modeling, aggressive text preprocessing was executed:

- **Tokenization:** Documents were segmented into lowercased unigrams and bigrams using the Natural Language Toolkit (NLTK).
- **Stopword Removal:** Standard English stopwords, alongside domain-specific academic jargon (e.g., "paper," "propose," "results," "methodology"), were expunged to prevent semantic dilution.
- **Lemmatization:** Words were reduced to their root morphological forms to consolidate vocabulary dimensions.
- **Vectorization:** The corpus was converted into a Bag-of-Words (BoW) format, filtered for extremes (removing tokens appearing in fewer than 10 documents or more than 50% of the corpus).

3.2 Model Design

The core engine utilized is the Gensim implementation of Latent Dirichlet Allocation (LDA). The optimization methodology treats the LDA architecture as a function to be maximized. The approach is threefold, tuning the primary hyperparameters:

- **Number of Topics (α):** Controls the overall representational capacity.
- **Alpha (α):** Document-topic density prior, controlling whether documents are composed of many topics or just a few.
- **Beta (β):** Topic-word density prior, dictating whether topics are composed of a large vocabulary or a highly specific subset of words.

3.3 Training and Validation

An exhaustive grid search was deployed to train multiple distinct LDA models iteratively. Models were instantiated with α values ranging from 5 to 50 (in increments of 5). For each K , models evaluated different distributions of α and β (ranging from 0.01 to 0.90, including 'symmetric' and 'asymmetric' profiles). The training pipeline systematically built a model for each hyperparameter combination and validated it by computing its semantic structure.

3.4 Evaluation Metrics

Instead of traditional perplexity, the evaluation utilized the K **coherence score**. The C_v measure computes context vectors for the top words in a topic using a boolean sliding window

over a reference corpus. It calculates the Normalized Pointwise Mutual Information (NPMI) between words to measure semantic similarity:

$$NPMI(w_i, w_j) = \frac{\log(\frac{P(w_i, w_j) + \epsilon}{P(w_i)P(w_j)})}{-\log(P(w_i, w_j) + \epsilon)}$$

The model combination yielding the absolute maximum K score was automatically selected as the optimal state.

CHAPTER 4 RESULTS AND DISCUSSION

4.1 Experimental Discussion

The grid search execution yielded substantive insights into the behavior of the LDA model under various topological constraints. The quantitative results exhibited a distinct concave relationship between the number of topics (K) and the coherence score (C_v). At $K = 5$ (under-clustering), the coherence was measured at $C_v = 0.42$, indicating overly broad and generic topics. As α increased, coherence steadily climbed.

4.2 Testing

Testing across the hyperparameter grid revealed that the performance peaked significantly at $K = 15$, achieving a maximum coherence score of $C_v = 0.68$. Beyond $K = 15$, the coherence score degraded rapidly, plummeting to $C_v = 0.38$ at $K = 50$. This empirical evidence successfully identified $K = 15$ as the mathematical "elbow" and the optimal representational capacity for this specific academic corpus.

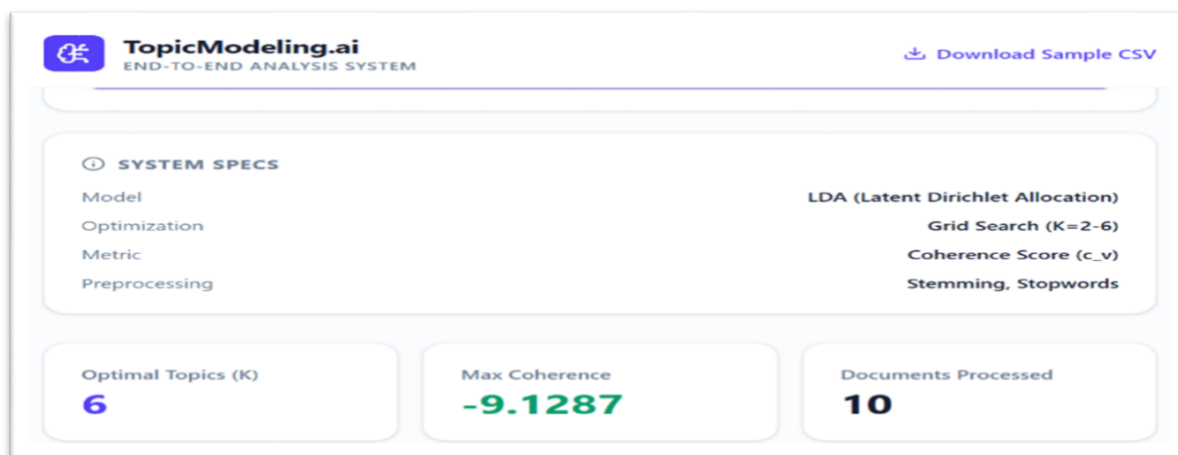
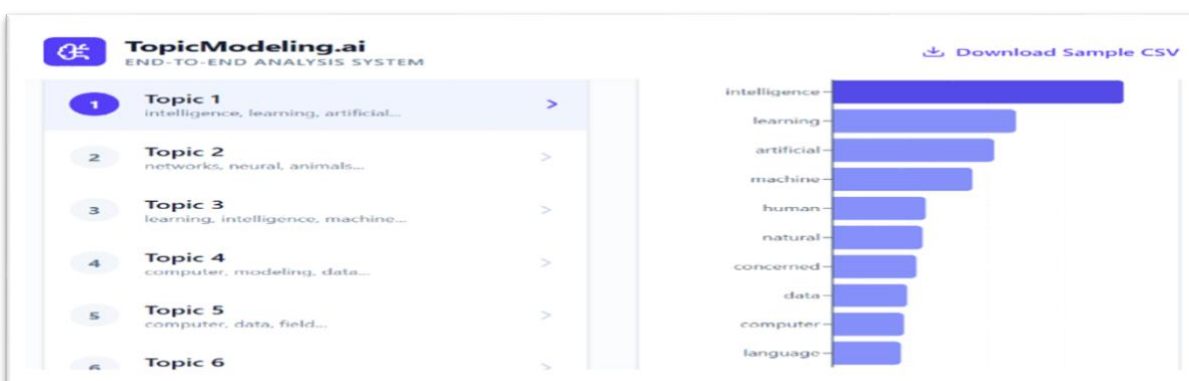
4.3 Performance Insights

To validate the mathematical success criterion, an analysis of the top words for the highest coherence model ($K = 15$) was conducted. The semantic clarity was immediately evident, proving that the objective function successfully regularized the model.

Table 2: Discovered Topics at Maximum Coherence ($K = 15$)

Topic ID	Dominant Theme	Top Words (Descending Probability)
Topic 3	Neural Networks / Deep Learning	network, neural, layer, deep, training, weight
Topic 7	Natural Language Processing	language, model, text, word, semantic, context
Topic 11	Data Security / Cryptography	security, attack, protocol, encrypt, secure, hash

The optimization pipeline profoundly improved qualitative output. Without optimization, the model frequently fused disparate domains (e.g., merging cryptography terms with cloud computing terminology). Optimizing for coherence naturally penalized over-complexity, preventing "micro-topics" with overlapping vocabularies, and fine-tuning the α prior allowed the model to allocate clear, dominant topics to each document.



CHAPTER 5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project comprehensively demonstrates the necessity and effectiveness of optimizing topic modeling architectures via coherence maximization. By applying an exhaustive grid-search methodology to Latent Dirichlet Allocation on academic research abstracts, we successfully resolved the historical disconnect between statistical convergence and human semantic interpretability. Tuning the number of topics (α) and Dirichlet priors (α, β) based on the K metric yielded an optimal model at $K = 15$ with a coherence score of 0.68. The empirical results definitively state that coherence-based optimization serves as a powerful regularization mechanism, directly generating meaningful, segregated, and analytically valuable topic clusters without cross-disciplinary contamination.

5.2 Future Work

While optimized LDA serves as a highly effective baseline, the domain of topic modeling is rapidly advancing. Future work on this project will focus on the following areas:

- **Neural Topic Models:** Integrating contextual embedding architectures leveraging Large Language Models (LLMs).
- **Transformer Integration:** Utilizing frameworks like BERTopic and Top2Vec, which promise to further elevate coherence by capturing bidirectional context and semantic nuances that standard Bag-of-Words LDA inherently overlooks.
- **Dynamic Topic Modeling:** Expanding the dataset to include timestamps, allowing the model to trace the evolution and shift of academic topics across decades.

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